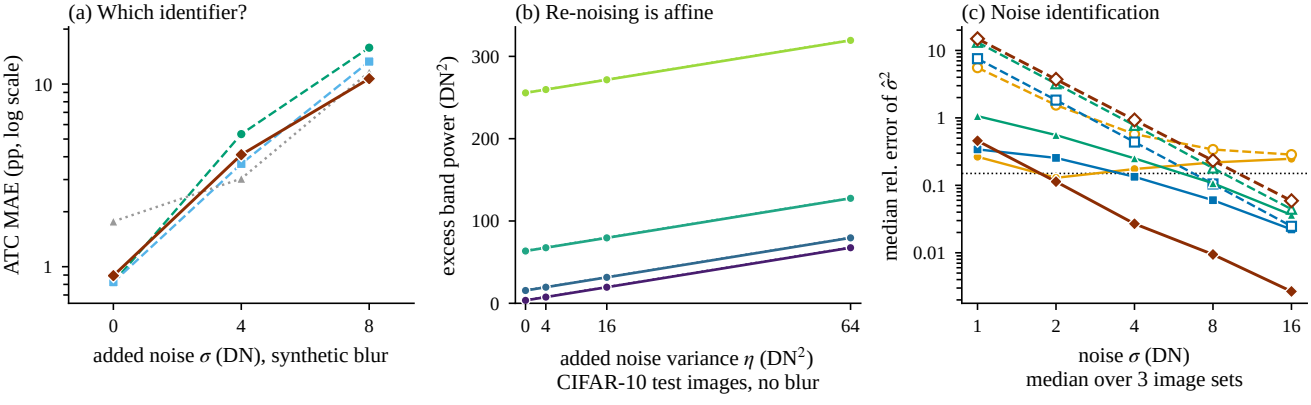
(a) Which identifier?
ATC MAE (pp, log scale)
10
1
0
4
8
added noise σ (DN), synthetic blur
(b) Re-noising is affine
excess band power (DN²)
300
200
100
0
0 4 16 64
added noise variance η (DN²)
CIFAR-10 test images, no blur
(c) Noise identification
median rel. error of σ̂²
10
1
0.1
0.01
1 2 4 8 16
noise σ (DN)
median over 3 image sets
(a) level from ŝ
(a) level from var. of Laplacian
(a) level from spectral slope
(a) level from feature regressor
(b) σ = 2 DN
(b) σ = 4 DN
(b) σ = 8 DN
(b) σ = 16 DN
(b) theory σ² + η
(c) re-noising (ours)
(c) v2 joint fit
(c) MAD (DCT band)
(c) Immerkær 1996
(c) dashed, open: with blur